HW-04 — Advanced DES: Inventory or Queue Extensions

Module M06 | Chapter 8 | Weight 10% | Difficulty ⋆⋆

Module: M06 Chapter: 8 Weight: 10% Due: End of Week 9

Overview

Build and analyze a simulation model that goes beyond the basic M/M/c queue. Choose one of two tracks. Both tracks require 30 replications, 95% confidence intervals, at least 3 scenario comparisons, and a professional memo.

1 Track A — (s, S) Inventory Control

System: A medical supply warehouse uses a continuous-review (s, S) inventory policy. Parameters:

Parameter	Value
Daily demand	Poisson($\lambda = 8$ units/day)
Lead time	Uniform[2, 5] days
Holding cost	$2/unit/day
Fixed ordering cost	$50/order
Shortage (backorder)	$20/unit/day
Simulation horizon	365 days

Tasks:

1. Implement the (s, S) model in SimPy (or using `simdes.models.inventory.SSInventory`).
2. Compare three policies: *Conservative* $(s = 30, S = 80)$, *Moderate* $(s = 20, S = 60)$, *Aggressive* $(s = 10, S = 40)$.
3. For each policy run 30 replications; report mean daily total cost and its 95% CI, decomposed into holding, ordering, and shortage components.
4. Use Common Random Numbers (CRN) to compare the best policy against the baseline $(s = 20, S = 60)$. Report the mean cost difference and its CI.
5. Plot the inventory sawtooth for one replication of each policy.
6. Write a 1-page memo recommending a policy to the warehouse manager.

2 Track B — Advanced Queue: Healthcare Triage Clinic

System: A triage clinic has three sequential stages: registration, triage nurse, and exam room. Arrivals follow a non-homogeneous Poisson process with morning peak ($\lambda = 12$/hr, 8–11am) and afternoon peak ($\lambda = 10$/hr, 1–4pm); off-peak $\lambda = 4$/hr.

Tasks:

1. Implement the three-stage model in SimPy.
2. Add a priority system: high-acuity patients ($p = 0.2$) skip the registration queue and go directly to triage.
3. Compare three staffing configurations (vary triage nurses $c \in \{1, 2, 3\}$).
4. Run 30 replications per configuration; report mean sojourn time with 95% CI for each acuity class separately.
5. Use CRN to compare the best configuration against $c = 1$.
6. Write a 1-page memo recommending a staffing level.

Grading Summary (both tracks)

Component	**Points**
Correct model implementation (no global state, seeds as args)	30
Sawtooth / time-series plot (labeled axes)	10
30 replications + 95% CI (all scenarios)	25
CRN comparison (paired, CI reported)	25
Professional memo (recommendation + CI + limitation)	10
Total	**100**

Full rubric: `rubrics/homework_rubric.pdf` | *Solutions: the instructor package (available to instructors on request).*